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Semantic segmentation in skin surface microscopic images with artifacts removal

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ABSTRACT

Skin surface imaging has been used to examine skin lesions with a microscope for over a century and is commonly known as epiluminescence microscopy, dermatoscopy, or dermoscopy. Skin surface microscopy has been recommended to reduce the necessity of biopsy. This imaging technique could improve the clinical diagnostic performance of pigmented skin lesions. Different imaging techniques are employed in dermatology to find diseases. Segmentation and classification are the two main steps in the examination. The classification performance is influenced by the algorithm employed in the segmentation procedure. The most difficult aspect of segmentation is getting rid of the unwanted artifacts. Many deep-learning models are being created to segment skin lesions. In this paper, an analysis of common artifacts is proposed to investigate the segmentation performance of deep learning models with skin surface microscopic images. The most prevalent artifacts in skin images are hair and dark corners. These artifacts can be observed in the majority of dermoscopy images captured through various imaging techniques. While hair detection and removal methods are common, the introduction of dark corner detection and removal represents a novel approach to skin lesion segmentation. A comprehensive analysis of this segmentation performance is assessed using the surface density of artifacts. Assessment of the PH2, ISIC 2017, and ISIC 2018 datasets demonstrates significant enhancements, as reflected by Dice coefficients rising to 93.49 (86.81), 85.86 (79.91), and 75.38 (51.28) respectively, upon artifact removal. These results underscore the pivotal significance of artifact removal techniques in amplifying the efficacy of deep-learning models for skin lesion segmentation.

1. Introduction

Skin is not only the largest organ in the human body but also the heaviest organ in the human body with 15% of the total weight of the body. Skin serves as the defense system from external threats and it reflects the health of the human body. The skin presents an amazing "theatre of life" in which the key concepts of biology and pathology can be observed, examined, and manipulated in real-time, unlike any other organs. Skin behaves as the mirror of a person's lifestyle and environmental conditions. Skin reflects the general health, internal sickness, social, ethnic, dietary habits, mental conditions, and profession of a person [1].

Skin surface microscopy is an imaging technique to examine the skin by means of a microscope. The instrument used for this has a standard operating microscope with a camera attached to a side arm. Skin surface microscopy has been recommended because its improved clinical diagnosis accuracy of pigmented lesions would lessen the requirement for biopsy. For reliable diagnosis of cutaneous pigmented lesions, histological analysis of biopsy specimens is still the gold standard [2]. There are normally no complications to a skin biopsy. If there is a risk of complications due to an allergy to anesthesia, infection, or tattoo, the practitioner should take adequate precautions. Sudden pain, hemorrhage, edema, and discomfort may be experienced after the skin biopsy [3]. Skin numbness at the biopsy site and puncture harm to nearby tissue are possible side effects. To avoid or limit difficulties, a comprehensive history and physical examination are required when conducting a skin biopsy. The biopsy site should be correctly identified and conducted using strict aseptic methods. The patient should be advised of any potential complications during the preoperative, intraoperative, and postoperative procedures [4].

Peter Borrelus pioneered "skin surface microscopy" in 1655, and Johan Christophorus Kolhaus followed eight years later. Saphier, who coined the word 'dermatoscopy' in 1920, provided the first extensive explanation of conceivable applications of skin surface microscopy. In the early twentieth century, Otfried Müller invented portable monocular and binocular microscopes for therapeutic use. Goldman described the utility of this approach in many dermatoses and skin malignancies in the 1950s. Franz Ehring and colleagues in Germany coined the term "vital histology of the skin" in 1952. Kreusch and Rassner

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Table 1
Different terminologies used Skin Surface Microscopy [5].

Term	Coined by
Dermatoscopy	Saphier
Surface microscopy	Soyer and colleagues
Incident light microscopy	Fritsch and colleagues
Epiluminescence light microscopy	Pehamberger and colleagues
Dermoscopy	Friedman and colleagues

invented a compact binocular stereo microscope in 1990 that was simple to operate and comfortable to handle. Videodermoscopy has been generally common now as a result of numerous advancements in digital camera technology, particularly the introduction of CCD cameras. Teledermoscopy has been made possible thanks to the popularity of the internet and incredible developments in computer technology [5]. Table 1 shows different terminologies used in skin surface microscopy.

Skin surface microscopy has been referred to by a variety of names in the research. Confusion about the terminology had been discussed in a consensus meeting organized by the Committee on Analytical Morphology of the Arbeitsgemeinschaft Dermatologische Forschung (ADF) on Nov. 17, 1989, in Hamburg. Surface microscopy or epiluminescence microscopy are suggested terms to describe the technology. The name dermatoscopy refers to the use of a dermatoscope, which is a lightweight device analogous to an otoscope [6]. The name is still a personal choice after centuries of evolution [7].

Table 5 shows a comprehensive overview of the historical evolution of skin imaging techniques. These techniques, spanning from epiluminescence microscopy to modern dermatoscopy and dermoscopy, have revolutionized dermatology by enabling detailed examination of skin lesions. However, despite these advancements, the presence of artifacts remains a persistent challenge throughout the study. These artifacts, such as hair and dark corners, commonly manifest in captured images, complicating the segmentation process and potentially affecting diagnostic accuracy. Therefore, our research endeavors to address this longstanding issue by proposing robust artifact removal techniques to enhance segmentation performance and ultimately improve the clinical utility of skin imaging technologies. Compared to other artifacts commonly found in dermoscopy images, such as hair and dark corners, these particular artifacts are predominantly inherited with skin images and can be referred to as "natural artifacts".

2. Related works

The main objective of our study is to increase accuracy in the skin lesion segmentation process by using artifact reduction. Hair detection and elimination techniques are widely used, but the proposal of identifying and removing dark corners represents a novel concept in our research. Deep learning became an effective approach in skin lesion segmentation and classification [8]. In this section, we provide a comprehensive overview of different deep-learning algorithms employed for semantic segmentation.

2.1. Deep learning models

Semantic segmentation of an image is defined as the segmentation by pixels where each pixel is labeled to a particular category. The introduction of U-Net architecture emerged as an important tool for semantic segmentation in biomedical imaging. This has implications for the study of skin lesion segmentation using dermoscopic images. Here U-Net semantic segmentation architecture is used to create the segmentation model to analyze the influence of artifacts. The U-Net architecture was initially developed by Olaf Ronneberger and his team to segment biomedical images [9].

The U-Net architecture looks like a U-shaped network that consists of two paths. Fig. 1 shows the architecture of U-Net. The first path on

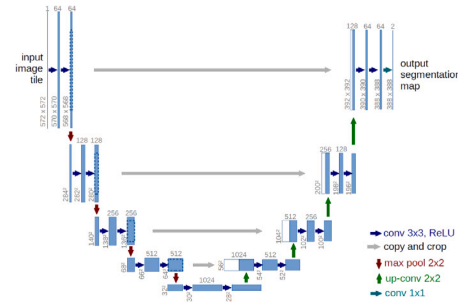


Fig. 1. U-Net Architecture [9].

the left side is a shrinking path. The second path on the right side is an expanding path. The main component of both paths is designed with Convolutional neural network (CNN) architecture. This approach establishes a local connection with every neuron. The connectivity is defined in terms of filter size. Here it is 3x3 with un-padded convolution. The activation function rectified linear unit (ReLU), is also incorporated with each block. After each block, a 2x2 max pooling operation with stride 2 diminishes the spatial dimension [10]. The aim of shrinking the path is to create convolutional layers with downsampling. The expansive path consists of convolutional layers that increase the feature channels with the upsampling operation. A 2x2 filter size is employed here for up-convolution. The feature map is increased by concatenating the feature map from the shrinking path. The pixels in the boundary may be lost. So a cropping function is essential. 1x1 convolution is employed in the final layer to map the feature vector to the ground truth labels. The entire network has 23 convolutional layers to map the three-dimensional matrix to the number of desired classes.

It is common to use the U-Net architecture itself to segment skin lesions in applications such as dermatology. Codella et al. [11] have taken advantage of this architecture to segment the ISIC 2016 dataset. Different color spaces were used in the empirical experiments in order to determine the results. The training set had also been expanded by employing data augmentation, and the input image's size had been set to 128x128 pixels. By employing non-linear distortions, augmentation effectively simulates the variations of soft-tissue biological structures. The tunable parameters are (i) the size of the kernel (ii) the number of convolution filters (iii) the standard deviation of the noise in each Gaussian noise layer (iv) the degree of dropout used (v) learning rate and (vi) momentum. The training data for segmentation for ISIC 2016 was split into two partitions of 80% for training and 20% for validation. The performance measures that they were able to obtain from the method: area under the receiver operating characteristic curve is 0.843, average precision is 0.649 and specificity is 36.8%. These results indicate that the model has a good ability to predict outcomes, which can indicate that the model could be improved in terms of its ability to differentiate between true and false positives.

D. Roja Ramani and S. Siva Ranjan [12] have implemented the U-Net architecture as part of the process of skin cancer classification that involved the segmentation of skin lesions. Here, the preprocessing was done using the fastICA approach, which lowers the amount of noise in the images. Diseases have been recognized by extracting several features using the Relevance Vector Machine (RVM) classifier. They carried out experiments on the ISBI 2016 dataset and performance metrics maximum Jaccard Coefficient, Dice Similarity Coefficient, Average Precision, and Accuracy achieved by the U-Net-based segmentation were 82.4%, 91.2%, 94.1%, and 93.5%, respectively.

Federico et al. [13] have used the U-Net architecture in their experiments. However, in order to improve the training set, they employed two types of Generative adversarial networks. The author first utilized Deep Convolutional GAN (DCGAN), a well-known type of GAN that employs deep convolutional neural networks and is widely used for image generation, to create the images. In contrast, Laplacian GAN (LAPGAN) uses a Laplacian pyramid approach to generate high-resolution images from low-resolution inputs. They have experimented in the ISIC 2017 dataset with image size 192 × 256 and obtained a Jaccard index of 0.717.

The U-Net design was improved by the researchers to enhance the effectiveness of skin lesion segmentation. Researchers have proposed several methods for skin lesion segmentation using deep learning models. Tang P et al. [14] have proposed a method based on the separable U-Net with stochastic weight averaging. The proposed Separable U-Net framework takes advantage of the separable convolutional block and U-Net architectures, which can extremely capture the context feature channel correlation and higher semantic feature information to enhance the pixel-level discriminative representation capability of fully convolutional networks (FCN). The performance of the aforementioned skin lesion segmentation methods was evaluated on several datasets, including the 2016 Skin Lesion Challenge (SLC) dataset from the International Skin Imaging Collaboration (ISIC), which achieved an average Dice coefficient of 93.03% and a Jaccard index of 89.25%. On the ISIC 2017 dataset, the average Dice coefficient and Jaccard index were 86.93% and 79.26%, respectively, while on the PH2 dataset, they were 94.13% and 89.40%, respectively.

Another approach proposed by Alom et al. [15] involves using recurrent U-Net models, which combine the power of U-Net, residual networks, and recurrent convolutional neural networks, for skin lesion segmentation. They evaluated their models on the ISIC 2016 dataset, reporting average dice coefficients of 0.96 and Jaccard indices of 0.93. Kamrul Hasan et al. [16] proposed a lightweight network called Dermoscopic Skin Network (DSNet) for robust skin lesion segmentation. They used depth-wise separable convolutions to project the learned discriminating features onto the pixel space at different stages of the encoder to reduce the number of parameters. They compared the performance of their method with U-Net and Fully Convolutional Network (FCN8s) models and reported mean Intersection over Union (mIoU) scores of 77.5% and 87.0% for ISIC-2017 and PH2 datasets, respectively.

Yading et al. [17] have proposed an automatic skin lesion segmentation algorithm with fully convolutional networks. This architecture consists of 19-layer deep convolutional neural networks and the training process was conducted with the loss function based on Jaccard distance. They conducted the experiment with the ISIC 2016 dataset and obtained the Jaccard Index is 84.7%. Zhang et al. [18] have proposed a multiscale network learning for skin cancer segmentation. The multiscale network is added to the conventional U-Net architecture. The proposed supervised deep network concatenates decoder layers to handle changes in the lesion sizes. The performance of the aforementioned skin lesion segmentation methods was evaluated on several datasets, including the 2016 Skin Lesion Challenge (SLC) dataset from the International Skin Imaging Collaboration (ISIC), which achieved an average Dice coefficient of 93.03% and a Jaccard index of 89.25%. On the ISIC 2017 dataset, the average Dice coefficient and Jaccard index were 86.93% and 79.26%, respectively, while on the PH2 dataset, they were 94.13% and 89.40%, respectively.

Ramadan et al. [19] introduced a novel approach that involves a multi-input color U-Net model (CU Net). The CU Net consists of three distinct models, each with varying numbers of input colors: the Single Input Color U-Net (SICU-Net), the Dual Input Color U-Net (DICU-Net), and the Triple Input Color U-Net (TICU-Net). These models provide an effective means of leveraging color information to improve segmentation accuracy and have the potential to greatly enhance the accuracy of skin lesion diagnosis in the future. To evaluate the effectiveness of their proposed CU Net model, the authors conducted experiments using the ISIC 2017 dataset, obtaining Jaccard Index values of 74.13%, 73.56%, and 74.88%, respectively, for the SICU-Net, DICU-Net, and TICU-Net models. These impressive results demonstrate the potential of the CU Net architecture to achieve high levels of accuracy in skin lesion segmentation tasks.

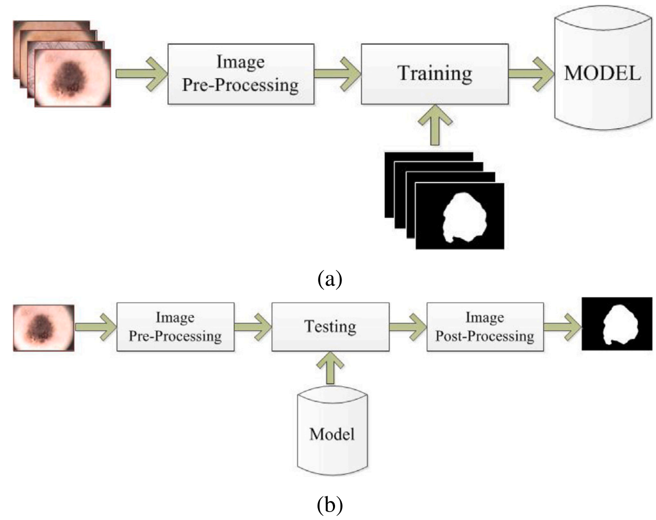


Fig. 2. Schematic diagram (a) Training (b) Testing and Analyzing.

Yutong et al. [20] proposed a mutual bootstrapping model for skin lesion segmentation and classification. This architecture consists of three Segmentation networks (i) Coarse segmentation network (coarse-SN) (ii) mask-guided classification network (mask-CN) (iii) enhanced segmentation network (enhanced-SN). They have experimented with the model with the ISIC 2017 dataset and PH2 dataset and obtained Jaccard index of 80.4% and 89.4%

2.2. Dark corner removal

Samuel William Pewton and Moi Hoon Yapr [21] proposed to address the underexplored Dark Corner Artifact (DCA) phenomenon within skin lesion image datasets. Through analysis of a curated dataset comprising 9,810 images, including 2,631 images with varying intensities of DCA, the study aims to enhance understanding of DCA occurrence and types. Additionally, the author introduces automated DCA detection and removal methods to improve dataset quality. The study investigates the effects of DCA removal on a binary classification task (melanoma vs. non-melanoma), suggesting potential benefits in shifting network activations towards skin lesions.

3. Proposed method

Deep learning models are constructed from what seems like an infinite number of simple computational elements. The microscopic description makes it crystal clear how a trained neural network assembles a function from these fundamental constituents. The description is put together using functions that may change input into output while following a process. However, the macroscopic description could not explain why a deep learning model computes the function. It is absurd to think that a few basic mathematical ideas will do the trick [22].

In this paper, an investigation of the performance of deep learning models is made with known features. For that one simple deep architecture is selected to segment the skin lesion image and evaluate the performance of the model with different pre-processing techniques. The pre-processing techniques are used to find and remove the natural artifacts present in the skin surface microscopic images. This approach also analyzes the possibility of post-processing techniques. Fig. 2 shows the schematic representation of training and testing.

3.1. Image acquisition

PH2 DATASET: The PH2 Database was created as a result of a collaboration between the Universidade do Porto, the Instituto Superior

Técnico Lisboa, and the Dermatology Service of Hospital Pedro Hispano in Matosinhos, Portugal. This library contains 200 dermoscopic images (8-bit RGB 768x560 pixels) captured with a Tuebinger Mole Analyzer system at a magnification of 20x and under the same conditions [23].

ISIC DATASET: The International Skin Imaging Collaboration (ISIC) is an academia-industry collaboration that introduces worldwide dermatologic imaging standards in melanoma research. ISIC conducts challenges in skin imaging by establishing a public repository of annotated high-quality skin images. ISIC 2016 dataset provides 900 dermoscopic lesions images for training and 370 images for testing. The ground truth masks of training and data are provided with this [24]. ISIC 2017 dataset contains 2000 dermoscopic lesions images for training and 600 images for testing. The ground truth masks of training data are provided with this [25].

3.2. Image pre-processing

The skin images contain different types of artifacts. The most common artifacts in PH2 and ISIC datasets are hair and dark corners. The presence of these two artifacts is seen in most of the raw images. Hair can be a common feature in dermoscopy images, particularly in areas with abundant hair growth, such as the scalp or beard region. Hair can cause shadows on the skin's surface, which can interfere with the visualization of underlying structures and potentially obscure important diagnostic features [26]. Dark corners or vignetting can be associated with microscope optics, specifically with the microscope objective lens. The objective lens is responsible for collecting and focusing the light that passes through the sample onto the imaging sensor or eyepiece. However, due to the design of some microscope objectives, light can be lost or scattered at the edges of the image, resulting in darker corners or vignetting. This can be particularly noticeable in microscopes with large numerical apertures or high magnification. Hair and dark corners in dermoscopy can be considered natural artifacts, as they are inherent to the imaging process and can occur even with well-aligned, properly illuminated instruments. but their presence or absence should be evaluated on a case-by-case basis and taken into consideration when interpreting the images [27].

3.2.1. Hair and ruler mark artifacts removal

Hair is an important protein filament for maintaining skin health. But in skin surface imaging, hair acts as a natural artifact that hides the information of skin. So the removal of hair is necessary for the visual examinations. The DullRazor method was developed to remove the hair from the dermoscopic images [28]. The ruler marks on the skin images have a similar structure to the hair follicles. Here a modified approach of the dull razor method is employed to remove the hair and ruler marks. The modified algorithm plays a pivotal role in the process of removing hair from skin images. Firstly, the image undergoes enhancement through contrast-limited Adaptive Histogram Equalization (CLAHE), a technique aimed at improving image quality by enhancing local contrast. Subsequently, morphological black hat operation is applied, which involves highlighting darker regions against a lighter background, thereby aiding in the detection and removal of hair. Following this, dilation further refines the process by expanding the highlighted areas, facilitating more precise hair removal. Finally, the procedure culminates in the reconstruction of the skin lesion using image in-painting techniques, as detailed in [29]. This comprehensive approach ensures effective hair removal while preserving the integrity and clarity of the skin image for subsequent analysis or diagnosis.

3.2.2. Dark Corner (DC) artifacts removal

Dark Corner is another artifact that is associated with the intrinsic part of a microscopic camera. These intrinsic artifacts can be removed easily from the images if all the skin lesion images contain a uniform dark corner region. But the dataset contains different types of dark corners and some of the images do not have dark corners. The dark corners cover the most portion of the skin images and resemble the skin abnormality. So the removal of dark corners is very essential to improve the classification of melanoma. Table 2 shows the classification of dark corners.

Table 2
Classification of dark corners.

Types of dark corners	Percentage of dark corners in the image
Small	1%–25%
Medium	26%–50%
Large	> 50 %
No DC	<1%

The dark corners can be classified according to the area covered, the number of corners present, symmetry, etc. Here the algorithm assumes that the dark corners are present in four corners and circular in shape. The image is divided into four segments to analyze the shape of the dark corners (Fig. 7 (b)). I_{RGB} represents the input color image.

The dark corner removal (DCR) algorithm is as follows:

Step 1: convert the input color image into a binary image with linear /adaptive thresholding. In the adaptive thresholding, the levels of grayscale image I_{GRAY} are calculated with the OTSU method. In the linear thresholding, a threshold value T_{SEG} is fixed. I_{BW} is the thresholded binary image.

Step 2: Find the presence of dark corners while segmenting the pixels of the four corners: Four small boxes are extracted from the four corners of the RGB image. The sample matrix called DC_X is used to determine whether there are any dark corners in the four corners (Left Top, Right Top, Left Bottom, and Right Bottom).

$$DC_{LT} = I_{RGB}[1 : P_{DC}, 1 : P_{DC}] \tag{1}$$

$$DC_{RT} = I_{RGB}[1 : P_{DC}, C_o - P_{DC} : P_{DC}] \tag{2}$$

$$DC_{LB} = I_{RGB}[R_o - P_{DC} : P_{DC}, 1 : P_{DC}] \tag{3}$$

$$DC_{RB} = I_{RGB}[R_o - P_{DC} : P_{DC}, C_o - P_{DC} : P_{DC}] \tag{4}$$

P_{DC} is the number of successive pixels in the DC matrix. R_o and C_o are the rows and columns of the input image. The presence of dark corners can be estimated by the average of DC matrices (M_{DC}) which should be less than T_{DC}.

$$M_{DC} = \frac{\sum DC_X}{P^2_{DC}} \tag{5}$$

T_{DC} is the optimum threshold value that determines the presence of a dark corner. The values of $T_{DC}$and P_{DC} are dependent. The presence of DC in each corner can be noted separately.

Step 3: Finding the circles using hough transform: This step finds the circular regions from the grayscale image using hough transformation and creates a list with the following parameters (i) centroid (ii) radius (iii) score. Score is the ratio of the number of active pixels in the locus of the circle to the perimeter of the circle. The parameter list will be sorted to find the largest or best circles with radius and the deviation of the centroid from the origin. D_{CENT} is the distance between the center points of the estimated circle to that of the image center.

$$D_{CENT} = \sqrt{(x-a)^2 + (y-b)^2} \tag{6}$$

Where (a,b) is the center point of the image and (x,y) is the center point of the predicted circle. The threshold value for the Maximum Distance T_{DCENT} should be greater than the estimated distance. The radius shall be in the range of T_{RMIN} and T_{RMAX}

$$T_{DCENT} > D_{CENT} \tag{7}$$

$$T_{RMIN} < R < T_{RMAX} \tag{8}$$

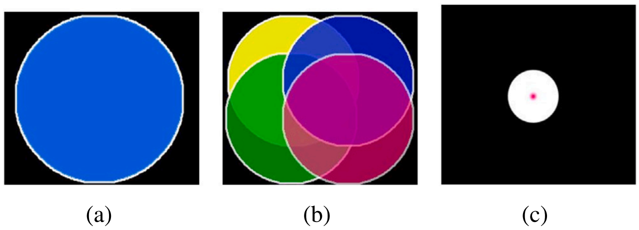


Fig. 3. Circle detection (a) uniform DC (b) Different DC (c) White regions indicate the location of the center of the detected circles.

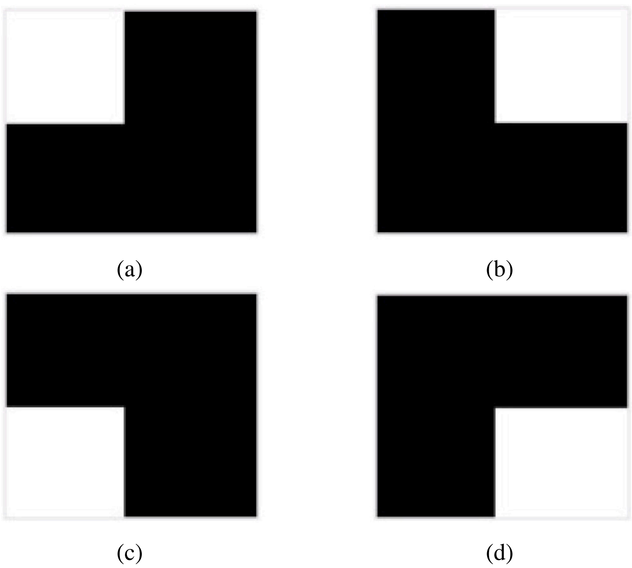


Fig. 4. Predefined masks for the DC (a) Left Top (b) right Top (c) Left Bottom (d) Right Bottom.

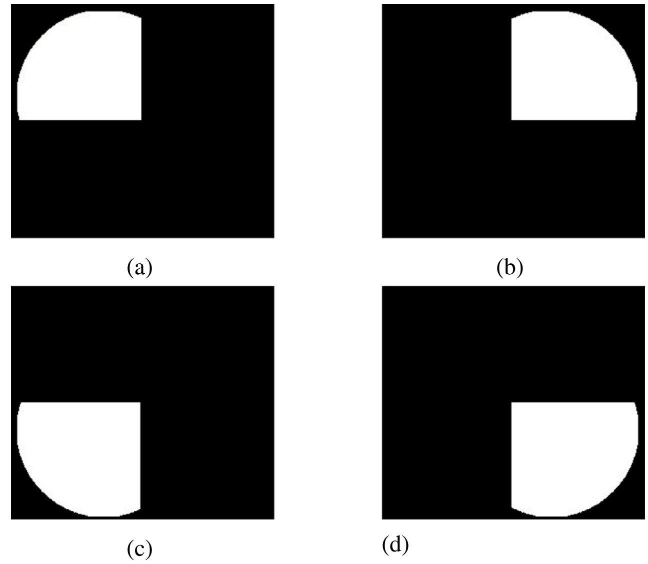


Fig. 5. Segmented masks for the DC (a) Left Top (b) right Top (c) Left Bottom (d) Right Bottom.

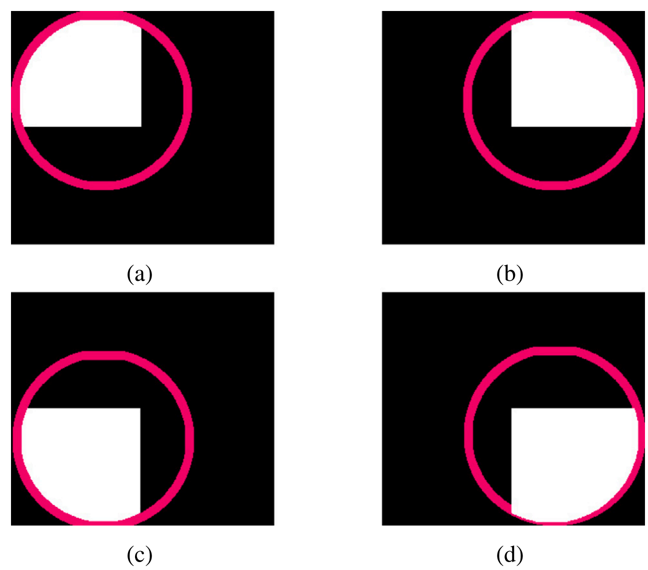


Fig. 6. Elliptical Fitting in the mask images (a) Left Top (b) Right Top (c) Left Bottom (d) Right Bottom.

If D_{CENT} of the largest best circle is approximately equal to zero, there will be a chance for four uniform dark corners. Here only one circle should have satisfied the conditions as shown in Fig. 3(a), if step 2 gives the presence of four dark corners. But this condition rarely occurs (Fig. 3(b)). Here all the circles that satisfy the condition in step 3 will be tested for DC. Fig. 3(c) represents the location of the best circles. For that four predefined masks as shown in Fig. 4 will be multiplied with the I_{BW} so that the masks in Fig. 5 may be obtained.

Step 4: Finding the curved shapes using elliptical fitting

This is an alternative approach to finding the curved shapes near the corners. The thresholded image will be segmented into four different images and the parameters with elliptical fitting. Fig. 6 shows the detection of the dark corner region with elliptical fitting. This predicts the largest elliptical or circular boundary and it returns the center, axes, radius, orientation, etc. Four different masks are created and combined to form one mask.

Step 5: Create dark corner masks: Step 3 and step 4 find the circular shapes in the corner. If these steps give better results, it will compare with step 2 for the number of dark corners present. A new mask is created by combining these four segmented masks as shown in Fig. 7(c). Another mask is created from this mask by filling the lines from the edges of the image to the boundary of the DC area as shown in Fig. 7(d). The lines may be drawn either horizontally, vertically, or both. Single precision masks are those made by drawing all the lines in a single direction. Double-precision masks are those made with lines extending in both directions.

If steps 3 & 4 find a circular region and step 2 cannot detect the presence of dark corners, step 2 will be processed with new parameters (T_{DCN} and P_{DCN}). If failed, the image will be considered as with no dark corner. If the presence of a dark corner is detected and the circular regions cannot be detected correctly with re-calculation, a default dark corner mask is assigned to the image. This case happens when the skin lesion resembles or overlaps with dark corners (figure 12(b)).

3.3. Deep learning model

The primary objective of our work is to analyze the impact of artifacts on skin lesion segmentation, particularly those arising from hair and dark corners. For this purpose, we have utilized the basic U-Net architecture, as depicted in Fig. 1 and described in Section 2.1.

3.4. Image post processing

Post processing consists of two methods (i) morphological operations and (ii) Dark corner removal. Morphological erosion removes pixels from the boundary of the image and morphological dilation adds

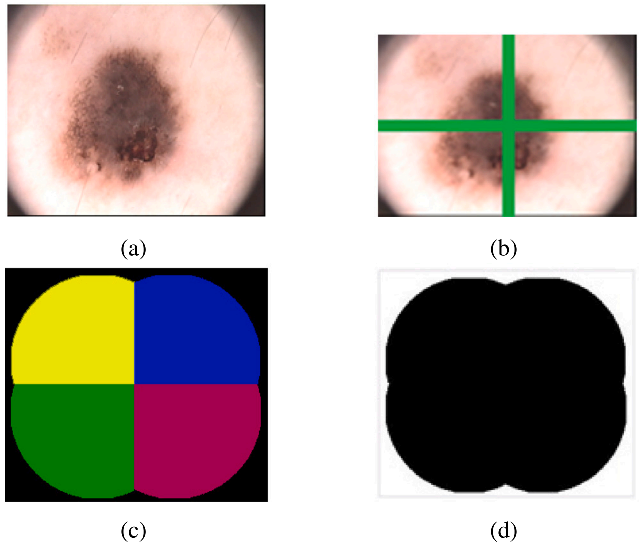


Fig. 7. (a) Original Image (b) divided into four segments for DC analysis (c) combined dark corner mask (d) complemented mask.

the pixels to the boundary of objects. Erosion eliminates scattered pixels and dilation fills the holes in the segmented region. The combined operation may result in small irregularities in the border of the region of interest. Here morphological closing and opening are proposed to enhance the quality of the segmented image.

The same image pre-processing methods can be used here to eliminate false predicted regions from the segmented mask. The binary mask is multiplied with a dark corner mask and pixels in the borders of the entire image will be removed. The dark corner mask is created with an algorithm described in section 3.2. The post-processing is here proposed for a transfer learning model where the trained model influences the natural artifacts of the trained dataset.

3.5. Surface density of artifact

The surface density of an artifact is defined as the two-dimensional mass of an artifact in an image per unit area [30]. It is the ratio of the number of white pixels to the total number of pixels in the corresponding artifact's mask. Surface Density (SD) can be calculated by

$$SD_{\text{artifacts}} = \frac{Area\ of\ the\ artifacts\ in\ pixels}{Total\ Area\ of\ the\ image} \tag{9}$$

Here two types of densities are extracted for analysis.

3.5.1. Hair density

Hair density is the ratio of the number of white pixels in the hair mask to the total number of pixels in the image.

$$HD_{\text{artifacts}} = \frac{Area\ of\ the\ Hair\ artifacts\ in\ pixels}{Total\ Area\ of\ the\ image} \tag{10}$$

3.5.2. Dark corner density

Dark Corner density is the ratio of the number of white pixels in the dark corner mask to the total number of pixels in the image.

$$DCD_{\text{artifacts}} = \frac{Area\ of\ the\ Dark\ Corner\ artifacts\ in\ pixels}{Total\ Area\ of\ the\ image} \tag{11}$$

4. Results and discussions

The PH2 dataset is taken into consideration when conducting studies to examine the impact of artifacts on the segmentation of skin lesions. This dataset contains only natural artifacts like hair and dark corners. Most of the images contain the hairs. Ruler marks are not present in the PH2 dataset, so ISIC dataset images are used to verify the algorithm that detects the ruler marks. Out of 200 images in the PH2 dataset, 187 images contain dark corners. Fig. 8 and Fig. 9 show the results of hair and ruler marker removal.

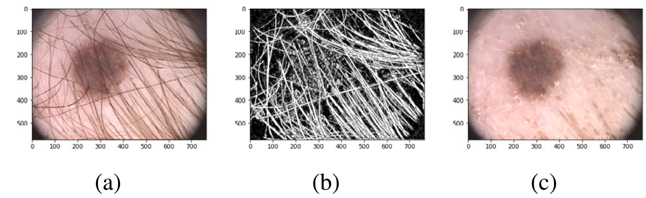


Fig. 8. Hair detection and removal (a) Input image (b) hair mask (c) reconstructed image [PH2 dataset].

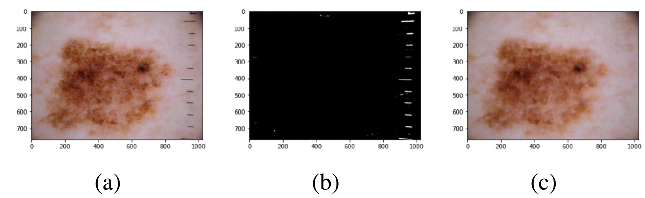


Fig. 9. Ruler mark detection and removal (a) Input image (b) ruler mask (c) reconstructed image [ISIC dataset].

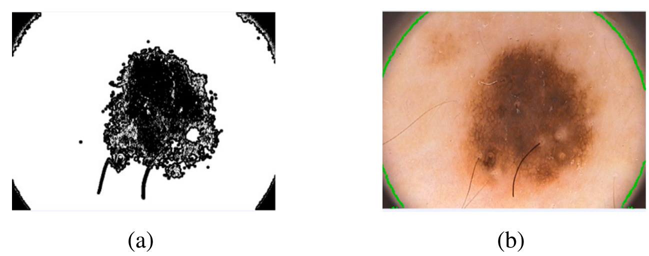


Fig. 10. Dark corner detection (a) Thresholded Image (b) Dark corner.

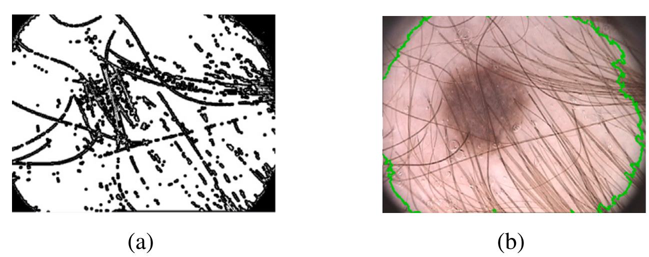


Fig. 11. Dark corner detection (a) Thresholded Image (b) Dark corner.

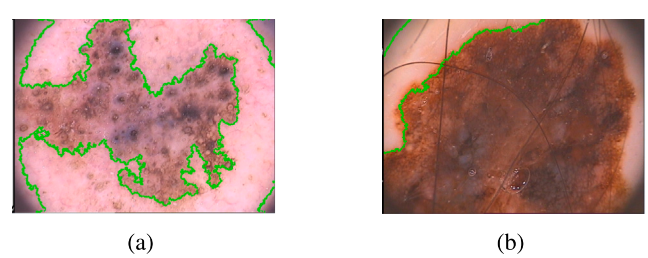
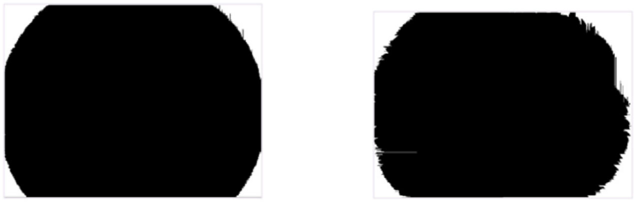


Fig. 12. Dark corner detection with active contours.

As per step 1 in the DC artifacts detection algorithm, the input image is thresholded with linear or adaptive thresholding. Here linear



(a) (b)

Fig. 13. Dark corner masks with (a) normal border (b) irregular border.

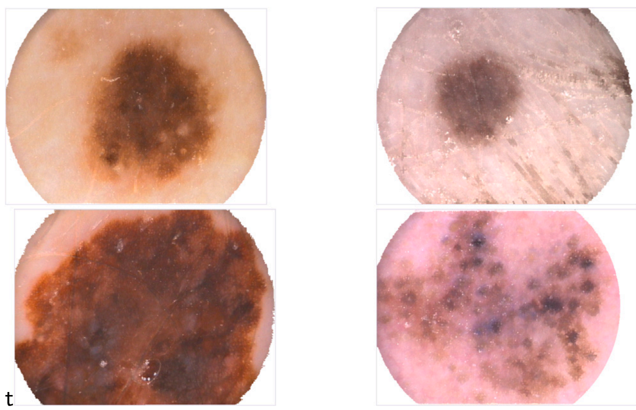


Fig. 14. Images with Dark Corner Artifact Removed.

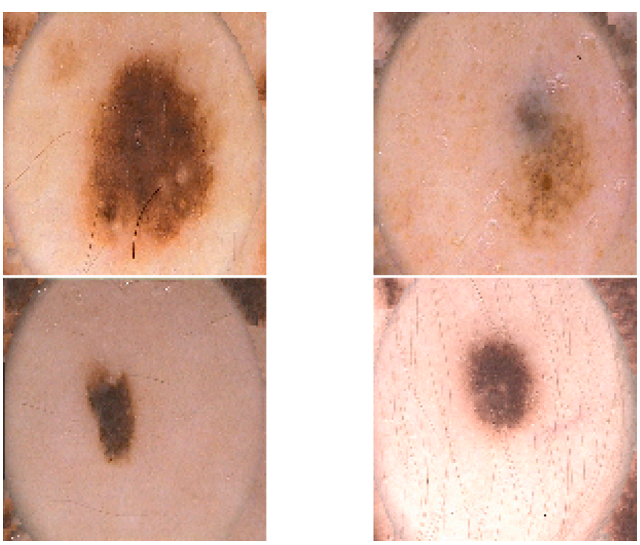


Fig. 15. Images with Dark Corner Artifact Removed Using Inpainting.

thresholding is used to segment the PH2 dataset. The second step is to find the presence of the dark corner by finding the image intensities in the corner. The third step of the algorithm consists of finding the number of large circles present in the thresholded image. Whenever it sees only one best circle, it assumes that this circle is the center of the DC. The threshold values for the best circle are as follows

$T_{DCENT}=300$ $T_{RMIN}=200$ $T_{RMAX}=300$

In the fourth step, the entire segmented image is divided into four parts and multiplied with different corner masks. The elliptical fitting method is applied to detect the elliptical/circular shape. This is an alternative approach to the previous step which may not detect the circular regions when multiple curved shapes are inscribed in the segmented image. Elliptical fitting will find the largest curve-shaped object with orientation, major and minor axes.

Figs. 10 and 11 show the dark corner detection of images. Fig. 10 shows the dark corner detection in a normal image which has a uniform dark corner detection. The dark corner is detected with the hair artifacts. In Fig. 11, the presence of hair gives dark corners with irregular borders. So the hair and ruler mark detection is proposed to be processed before the dark corner analysis.

Examples of dark corner identification using active contours are shown in Fig. 12. Fig. 13 shows the dark corner with a normal border and an irregular border. These results highlight deceptive dark corners. These kinds of images are uncommon. The number of circles detected with the algorithm shows ambiguity in the regions of dark corners. Our suggested method accurately recognizes the presence of a dark corner in most images. To prevent errors in identifying the precise location of the dark corner, a default dark corner mask is proposed for these images. Fig. 14 shows the DC-masked images with white filling and Fig. 15 shows the in-painted DC-removed images.

The algorithm for artifact removal was developed in MATLAB 2021 and Python. We trained our deep learning model using PyTorch on an NVIDIA Tesla V100 PCIe 32 GB GPU with CUDA support. The GPU was equipped with a GV100GL chip. We used the cuDNN library version 10.2 for accelerated deep-learning computations. The experiments were conducted to generate training models by changing the pre-processing conditions, aspect ratio, hyperparameters, and augmentation techniques. Here 140 images were randomly selected for training and 30 images were chosen for validation. The remaining 30 images were used for testing. The number of training data is increased with data augmentation methods. Images were flipped both horizontally and vertically, producing an array of images. Also, the images obtained using hair removal, dark corner removal, and inpainted dark corner algorithms were included with this.

The objective of this experiment is to analyze the influence of artifacts on deep learning models when created using U-Net as a learning method. The experiments are conducted with the following three circumstances (i) Without Pre-processing (WOP) (ii) Hair Removed (HR) (iii) DC Removed (DCR). In the first case, a training model is created with U-Net architecture without any artifact removal. In the second case, the training model was created after processing the hair and ruler mark removal algorithm. In the third case, the training model is created after removing the natural artifacts. For this, the dataset is organized into three groups, and the three training models were tested. Figs. 16–19 shows the resulting images of the proposed methods.

Tables 3–5 demonstrates that the model made without hair performs better than other models. The images that do not have the dark corner perform better as well. The experiments were started to create a training model without image augmentation. K-fold cross validation was

Table 3
Analysis of Deep Learning Model Performance: Training PH2 Data Without Pre-processing.

Test images	Accuracy (in %)	Sensitivity (in %)	Specificity (in %)	Dice Coeff. (in %)	Jaccard Index (in %)
WOP	93.39	89.44	92.08	81.37	71.02
HR	93.74	92.39	92.00	83.08	73.08
DCR	93.68	92.71	91.74	83.15	73.11

Table 4
Analysis of Deep Learning Model Performance: Training PH2 Data With hair removal.

Test images	Accuracy (in %)	Sensitivity (in %)	Specificity (in %)	Dice Coeff. (in %)	Jaccard Index (in %)
WOP	95.21	86.00	96.98	86.81	78.70
HR	**97.36**	**94.34**	**96.49**	**93.49**	**89.05**
DCR	**96.25**	**90.99**	**95.72**	**91.45**	**85.67**

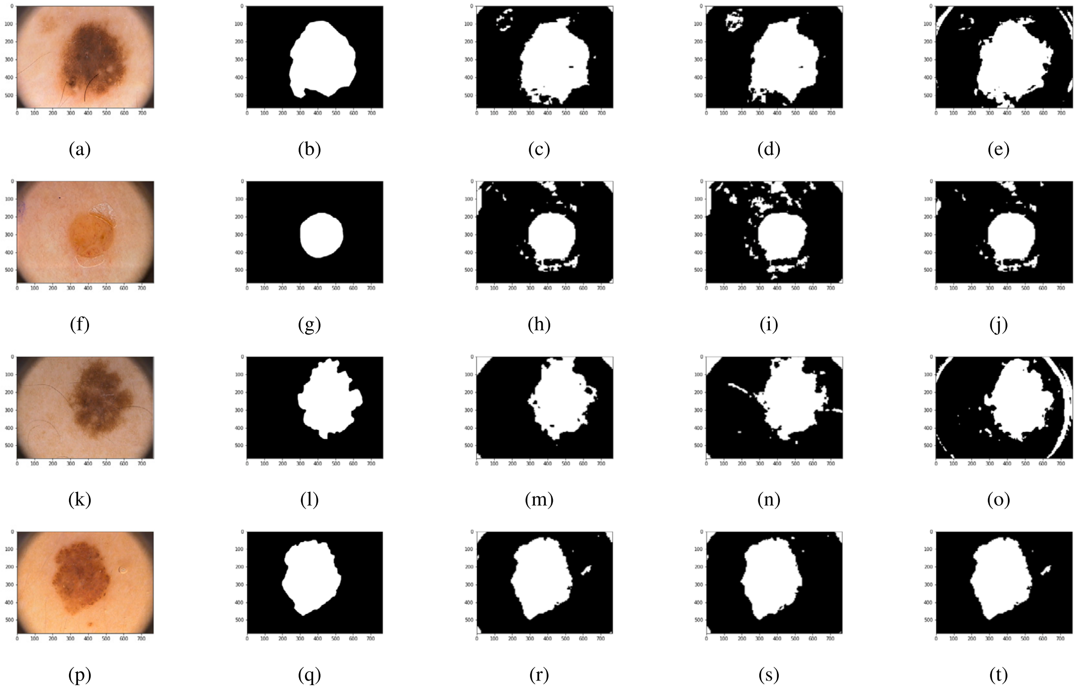


Fig. 16. Results of training model created without any pre-processing (a)(f)(k)(p) Original Image (b)(g)(l)(q) Ground truth mask (c)(h)(m)(r) Segmented testing image with hair removal (d)(i)(n)(s) Segmented testing image without pre-processing (e)(j)(o)(t) Segmented testing image with dark corner removal.

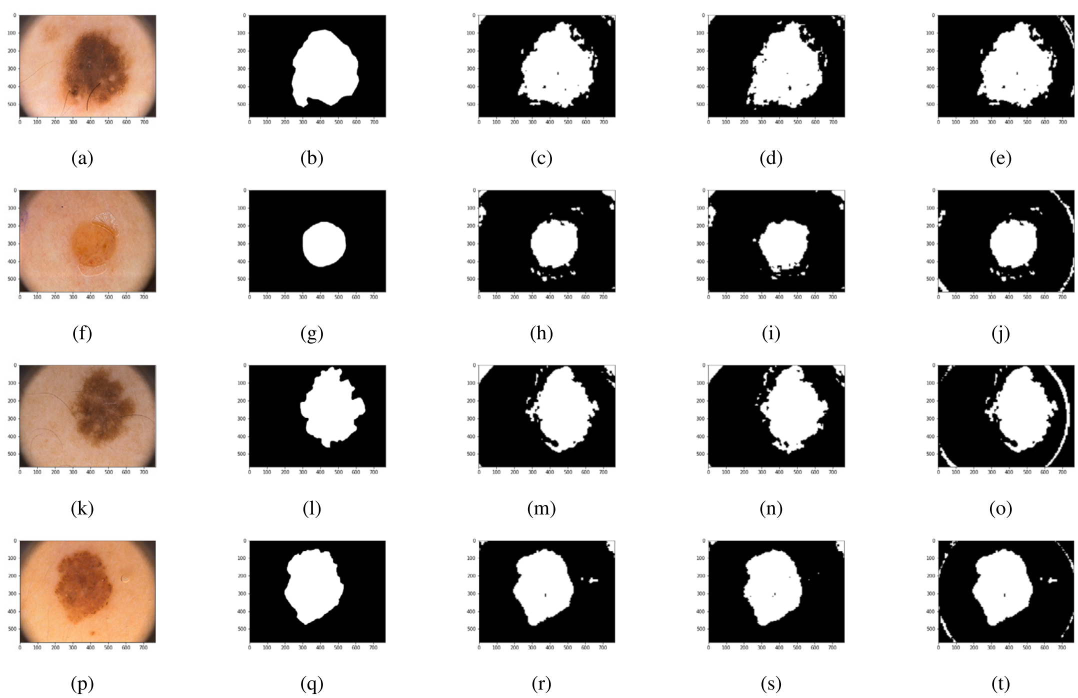


Fig. 17. Results of training model created with dark corner removal (a)(f)(k)(p) Original Image (b)(g)(l)(q) Ground truth mask (c)(h)(m)(r) Segmented testing image with hair removal (d)(i)(n)(s) Segmented testing image without pre-processing (e)(j)(o)(t) Segmented testing image with dark corner removal.

used to analyze the training model's performance. The various results demonstrated that the training procedure without pre-processing is equivalent to artifact removal in some cases. In contrast to WOP, negative effects with artifact removal were not observed. In order to improve the outcomes, the augmentation techniques were used with maximum epochs.

Fig. 20 shows the relation between the performance of each model and with density of artifacts. The graph is plotted with the performance of the entire PH2 dataset. This experiment's goal is to evaluate how well the training model performed when the corresponding artifacts were removed. So it is clear that hair artifacts can only be removed to train and test the images compared to raw images. The model created without dark corners shows better results for testing raw images. Table 6 shows its performance comparison with existing works.

Table 7 shows the performance comparison of the deep learning model created with the ISIC 2017 challenge dataset and tested

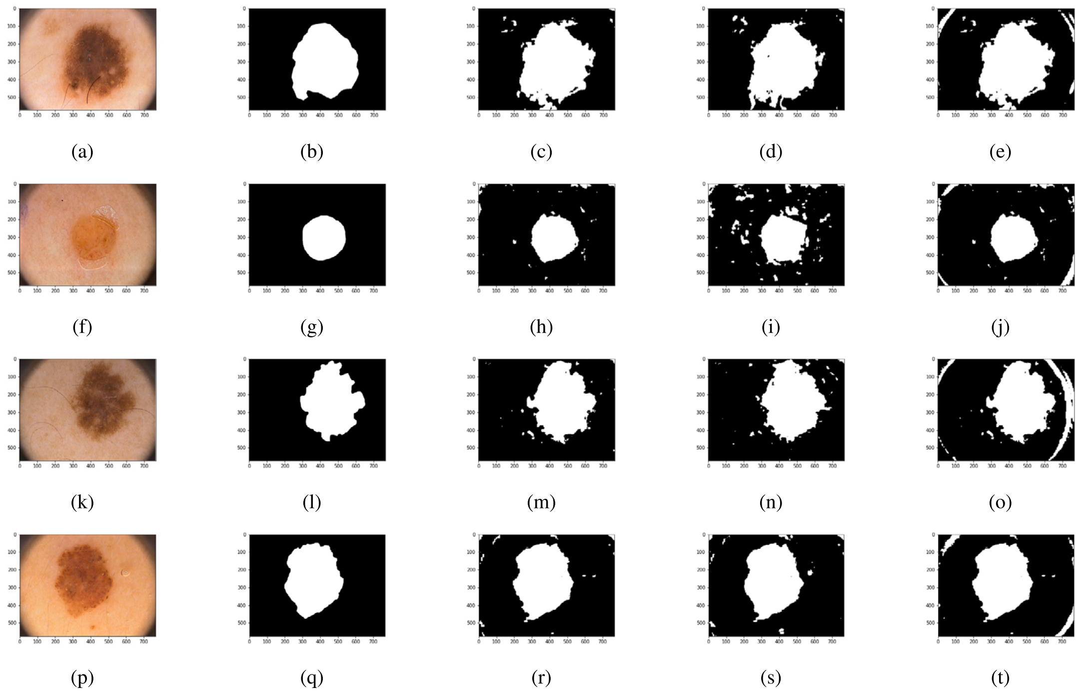


Fig. 18. Results of training model created with hair removal (a)(f)(k)(p) Original Image (b)(g)(l)(q) Ground truth mask (c)(h)(m)(r) Segmented testing image with hair removal (d)(i)(m)(s) Segmented testing image without pre-processing (e)(j)(n)(t) Segmented testing image with dark corner removal.

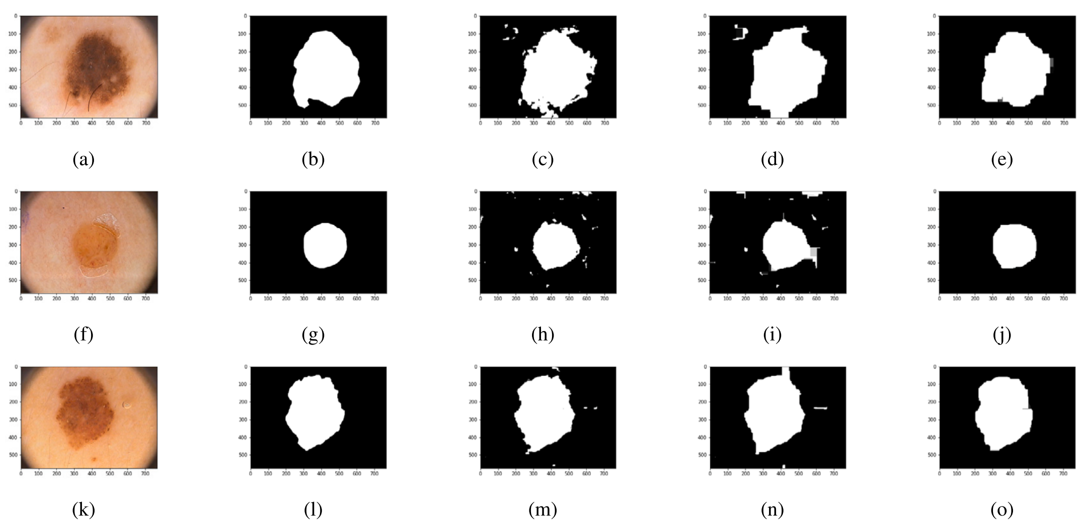


Fig. 19. Results of post-processing (a)(f)(k)(p) Original Images (b)(g)(l)(q) Ground truth mask (c)(h)(m)(r) DC removed (d)(i)(n)(s)Morphological close (e)(j)(o)(t) Morphological open.

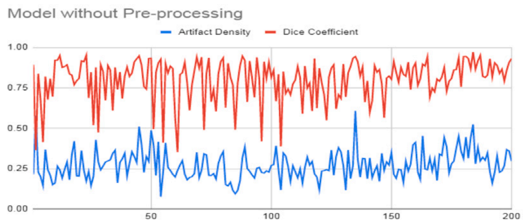
Table 5
Analysis of Deep Learning Model Performance: Training PH2 Data with dark corner removal.

Test images	Accuracy (in %)	Sensitivity (in %)	Specificity (in %)	Dice Coeff. (in %)	Jaccard Index (in %)
WOP	93.77	89.88	92.21	83.03	72.90
HR	95.22	89.57	94.69	87.07	79.68
DCR	92.92	88.12	91.48	81.93	71.30

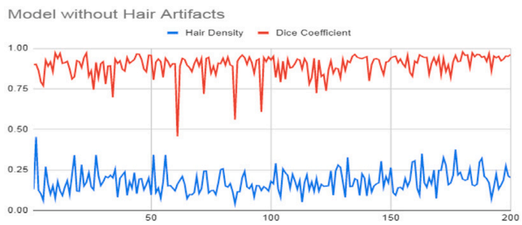
with different pre-processing conditions. This dataset contains 2000 training images and 600 testing images. The number of dark corner images was less in the dataset. Therefore, both images with artifacts eliminated were used for training. We used the PH2 dataset along

Table 6
Comparative Evaluation of Our Approach with Various Segmentation Algorithms Trained on the PH2 Dataset.

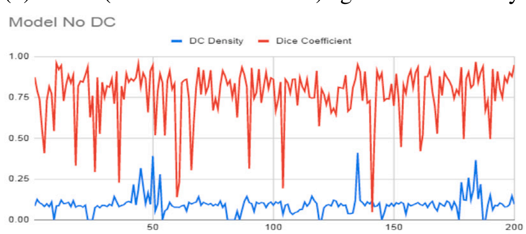
Method	Accuracy (in %)	Sensitivity (in %)	Specificity (in %)	Dice Coeff. (in %)	Jaccard Index (in %)
FCN [31]	92.82	90.30	94.02	89.03	80.22
U-Net [32]	92.55	81.63	97.76	87.61	77.95
Segnet [33]	93.36	86.53	96.61	89.36	80.77
SICU-Net [19]	91.75	92.78	91.25	87.87	78.37
TICU-Net [19]	93.16	89.70	94.80	89.42	80.87
Hair Removed	**97.36**	**94.34**	**96.49**	**93.49**	**89.05**
DC Removed	**96.25**	**90.99**	**95.72**	**91.45**	**85.67**



(a) model (without pre-processing) against natural artifacts



(b) model (without Hair artifacts) against Hair density



(c) model (without Dark corners) against DC density

Fig. 20. The graphs illustrate the Dice performance and surface density of artifacts of individual images in the PH2 dataset, with the training, validation, and testing sets distributed in a ratio of 70:15:15.

Table 7
Analysis of Deep Learning Model Performance: Training 2017 dataset created with Dark corner removal.

Test images	Accuracy (in %)	Sensitivity (in %)	Specificity (in %)	Dice Coeff. (in %)	Jaccard Index (in %)
WOP	86.72	73.45	85.73	79.91	68.28
HR	**92.53**	**88.68**	**93.54**	**83.49**	**79.28**
DCR	**96.22**	**89.07**	**96.53**	**85.86**	**80.48**

Table 8
Comparative Evaluation of Our Approach with Various Segmentation Algorithms Trained on the ISIC 2017 Dataset.

Method	Accuracy (in %)	Sensitivity (in %)	Specificity (in %)	Dice Coeff. (in %)	Jaccard Index (in %)
U-Net [32]	91.64	81.72	96.80	81.59	72.34
ASCU-Net [34]	92.6	82.5	96.5	83.0	74.2
SegAN [35]	94.10	–	–	86.70	78.50
DSM [36]	92.86	83.72	96.58	84.15	75.72
ERU [37]	–	–	–	86.1	78.1
SESV-FPN [38]	94.18	85.07	96.57	86.29	78.19
SLS-Deep [39]	93.60	81.60	**98.30**	**87.80**	78.20
SICU-Net [19]	92.63	86.37	94.65	85.15	74.13
DICU-Net [19]	92.49	85.38	94.79	84.77	73.56
TICU-Net [19]	93.13	83.64	96.21	85.63	74.88
Proposed	**96.22**	**89.07**	96.53	85.86	**80.48**

with the provided validation set. This deep learning model's output demonstrates improved U-Net architecture performance. Table 8 shows the performance comparison of the U-Net training model with artifacts removal in the ISIC 2017 test dataset, as compared to other algorithms.

Table 9
Analysis of Deep Learning Model Performance: Training 2018 dataset created with Dark corner removal.

Test images 2017 Full	Accuracy (in %)	Sensitivity (in %)	Specificity (in %)	Dice Coeff. (in %)	Jaccard Index (in %)
WOP	89.37	76.85	89.63	77.71	65.55
HR	**97.93**	**92.21**	**98.68**	**92.87**	**88.70**
DCR	**98.21**	**93.91**	**98.77**	**93.97**	**90.73**

Table 10
Analysis of Deep Learning Model Performance: Training 2018 dataset created with Dark corner removal.

Test images 2018 Test data	Accuracy (in %)	Sensitivity (in %)	Specificity (in %)	Dice Coeff. (in %)	Jaccard Index (in %)
WOP	76.52	60.14	77.76	51.28	37.93
HR	**87.65**	**81.45**	**88.67**	**72.11**	**67.70**
DCR	**88.32**	**85.69**	**87.29**	**75.38**	**71.63**

Our proposed model demonstrates superior performance compared to more sophisticated deep learning models, achieving either better results or satisfactory outcomes in various evaluation metrics and benchmarks. Tables 9–10 shows the performance comparison of deep learning models created with the 2018 dataset and tested with different datasets in different pre-processing conditions. The ISIC 2018 challenge dataset includes 2594 training images and 1000 testing images. Table 9 depicts the performance of a training model trained on the ISIC 2018 dataset and evaluated using full images from the ISIC 2017 dataset. Table 10 illustrates the model's performance when tested with the ISIC 2018 challenge test data.

5. Conclusion and future scope

Skin lesion segmentation is a crucial step in automated skin disease diagnosis and treatment monitoring. Traditional image processing techniques, machine learning algorithms, and deep learning approaches are all viable methods for skin lesion segmentation, with deep learning methods showing promising results in recent years. Here deep learning models are created with three different pre-processing conditions with basic U-Net architecture. This analysis shows that a better model can be created with artifact removal. The hair artifacts have a significant impact on segmenting the skin lesion images. Compared to dark corner artifacts, hair artifacts occur in the region of interest and overlap with other regions. The connected components in the hair artifacts split the skin pigment region irregularly. This will result in false detection. The removal of both artifacts is necessary to improve the performance of the U-Net Model. This analysis also gives a measure of the performance of conventional U-Net architecture. Removing artifacts is not expensive compared to developing a sophisticated model. Dark corner detection is a novel algorithm to analyze the segmentation model and it can be used with different datasets also. The total accuracy and dependability of dermatological diagnostic techniques could have been improved by this new finding.

The methodology helps improve the quality of dermoscopic image interpretation by systematically addressing and minimizing artifacts, providing a stronger basis for clinical decision-making. We expect a favorable knock-on impact on the larger field of dermatological research and patient care as we continue to hone and incorporate such technological innovations. This accomplishment highlights the possibility for further innovation in the discipline and represents a critical step forward in using deep learning techniques to enhance dermatoscopic assessments.

CRediT authorship contribution statement

Aneesh R.P.: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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